

INST414 Sprint 3 Presentation Outline

Slide 1: Title Slide

Title: Predicting Telecom Customer Churn Using Machine Learning

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Course: INST 414, Spring 2025

Date: April 27, 2025

Slide 2: Problem & Why It Matters

- Telecom companies lose money when customers cancel their service (churn)
 - Getting new customers costs more than keeping current ones
 - Predicting churn early helps companies act fast
 - Solving this helps improve customer happiness and reduce business loss
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Slide 3: Project Goal & Method

Goal: Use machine learning to predict which customers will leave

Data: Kaggle Telco Churn dataset (includes customer info and usage)

Steps I Took:

- Cleaned and prepared data
 - Split data into training and testing sets
 - Tried 3 models: Logistic Regression, Decision Tree, Random Forest
 - Tuned parameters with GridSearchCV
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Slide 4: Model Results

Model Accuracy Precision Recall F1 Score

Logistic Regression 79% 68% 52% 59%

Decision Tree 80% 71% 58% 64%

Random Forest 83% 75% 61% 67%

Best Model: Random Forest

- Gave strongest balance of results
 - Helped reduce wrong predictions on who will churn
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Slide 5: Why This Helps Business

- Helps company **find high-risk customers early**
 - They can offer **discounts or support** to keep those customers
 - Focus on people with **month-to-month contracts or no tech support**
 - Even **small drop in churn = big money saved**
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Slide 6: What's Next & What I Learned

Next Steps:

- Try better models like XGBoost or LightGBM
- Add more data like support tickets or satisfaction ratings
- Improve fairness and retrain the model regularly

What I Gained:

- Learned how to clean and model real data

- Saw how ML can solve real-world business problems
 - Learned to compare models and improve results
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Questions and Feedback?

I am happy to explain anything or receive feedback about what I would change next time.